

Lecture 1: Real Numbers

Mathematical Analysis

1 Numbers

Definition 1.1 (Natural Numbers). [def:natural-number] The *natural numbers* are the positive integers:

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

Convention 1.1 (Natural Numbers Exclude Zero). [conv:natural-numbers-exclude-zero] $0 \notin \mathbb{N}$.

Definition 1.2 (Integers). [def:integer] The *integers* are the whole numbers, including the positive integers, the negative integers, and zero:

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

Definition 1.3 (Rational Numbers). [def:rational-numbers] A real number is called *rational* if it can be expressed as a quotient of two integers. The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Proposition 1.1 (Unique Reduced Form). [prop:unique-reduced-rational-form] For every $x \in \mathbb{Q}$, there exists a unique pair $p, q \in \mathbb{Z}$ such that

$$x = \frac{p}{q}, \quad q > 0, \quad \gcd(|p|, q) = 1.$$

Remark 1.1. [rem:number-set-inclusions] Under the usual identifications, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$.

2 Fields

Definition 2.1 (Closed Operation). [def:closed-operation] An operation $*$ is said to be *closed* on a set A if for all $a, b \in A$, we have $a * b \in A$.

Definition 2.2 (Field). [def:field] A *field* is a set F equipped with addition and multiplication such that both operations are multiplication are

- closed operations on F ,
- associative, meaning for all $a, b, c \in F$, we have $(a+b)+c = a+(b+c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$,
- commutative, meaning for all $a, b \in F$, we have $a + b = b + a$ and $a \cdot b = b \cdot a$,
- have identity elements 0 and 1,
- each element has an additive inverse, such that for each $a \in F$, there exists $-a \in F$ with $a + (-a) = 0$,
- each non-zero element has a multiplicative inverse, such that for each $a \in F$ with $a \neq 0$, there exists $a^{-1} \in F$ with $a \cdot a^{-1} = 1$

Proposition 2.1 ($\mathbb{Q}$ Is a Field). [prop:rational-numbers-are-a-field] The rational numbers with their usual addition and multiplication form a field.

Remark 2.1 ($\mathbb{N}$ and $\mathbb{Z}$ Are Not Fields). [rem:natural-and-integers-not-fields] The natural numbers and the integers are not fields because they do not have multiplicative inverses for every non-zero element.

For a counter example, consider the integer 2. There is no integer x such that $2 \cdot x = 1$

3 Ordered Set

Definition 3.1 (Ordered Set). [def:ordered-set] A set F is *ordered* if it is transitive and reflexive and

$$\forall x, y \in F, \text{ either one of the following happens } x \leq y, x = y, \text{ or } y \leq x$$

It is also called partial ordered.

Definition 3.2 (Transitive). [def:transitive] An ordering on a set is *transitive* if

$$\forall x, y, z \in F, \text{ if } x \leq y \text{ and } y \leq z, \text{ then } x \leq z$$

Definition 3.3 (Reflexive). [def:reflexive] An ordering on a set is *reflexive* if

$$\forall x \in F, x = x$$

4 Irrational Numbers

Lemma 4.1 (Parity of a Square). [lem:square-parity] For every $n \in \mathbb{Z}$,

$$n^2 \text{ is even} \iff n \text{ is even},$$

and equivalently,

$$n^2 \text{ is odd} \iff n \text{ is odd}.$$

Proposition 4.1 (No Rational Square Root of 2). [prop:no-rational-square-root-of-two] There is no $q \in \mathbb{Q}$ such that $q^2 = 2$.

Proof. Suppose, for contradiction, that some $q \in \mathbb{Q}$ satisfies $q^2 = 2$. According to the proposition 1.1, and $q \in \mathbb{Q}$

$$\exists p, r \in \mathbb{Z}, q = \frac{p}{r}, r > 0, \gcd(|p|, r) = 1$$

Then we have

$$\left(\frac{p}{r}\right)^2 = 2 \implies p^2 = 2r^2$$

Since p^2 is even, the square-parity lemma implies that p is even. Therefore,

$$\exists k \in \mathbb{Z}, p = 2k.$$

So,

$$(2k)^2 = 2r^2 \implies 2k^2 = r^2$$

By the same argument, r is also even. Meaning that both p and r are even, which contradicts the assumption that $\gcd(|p|, r) = 1$.

So, it contradicts the assumption that $q \in \mathbb{Q}$ satisfies $q^2 = 2$. Therefore, there is no $q \in \mathbb{Q}$ such that $q^2 = 2$. $\square$

Remark 4.1 (Larger Number System). [rem:sqrt-two-larger-number-system] Within $\mathbb{Q}$, the proposition only says that the equation $q^2 = 2$ has no solution $q \in \mathbb{Q}$. Writing $\sqrt{2} \notin \mathbb{Q}$ presupposes a larger number system in which $\sqrt{2}$ exists.

Remark 4.2 (Completeness). [rem:rational-incompleteness] The existence, in a larger number system, of some $x \notin \mathbb{Q}$ satisfying $x^2 = 2$ does not by itself prove that $\mathbb{Q}$ is incomplete. Incompleteness must be demonstrated internally [TODO Pf link to completeness axiom].

Convention 4.1 (Irrational Numbers). [conv:irrational-numbers] We use $\mathbb{I}$ to denote the set of *irrational numbers*

Remark 4.3 (Number Sets). [rem:number-sets-inclusions-real] The set of real numbers $\mathbb{R}$ is the union of the set of rational numbers $\mathbb{Q}$ and the set of irrational numbers $\mathbb{I}$:

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$$

Also, in normal conventions, we have:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Proposition 4.2 (Integer–Irrational Root Dichotomy). [prop:nth-root-dichotomy] For $x, n \in \mathbb{N}$ with $n > 1$, $\sqrt[n]{x} \in \mathbb{Z}$ if x is a perfect n -th power, and otherwise $\sqrt[n]{x} \in \mathbb{I}$.

Remark 4.4 (Sources of Irrational). [rem:sources-of-irrational]

- By proposition 4.2, $\sqrt[n]{x}$ is irrational whenever $x \in \mathbb{N}$, $n > 1$, and x is not a perfect n -th power.
- Taking logarithms
- Constants defined by taking limits, such as π and e .

Remark 4.5 (Irrational Numbers Are Not Closed). [rem:irrational-not-closed] The irrational numbers are not closed under addition, multiplication, or exponentiation. For example,

$$\sqrt{2} + (-\sqrt{2}) = 0, \quad \sqrt{2} \cdot \sqrt{2} = 2, \quad (\sqrt{3})^{\log_3 4} = 2,$$

where all the inputs are irrational. Although $\pi, e \in \mathbb{I}$, it is not known whether either $\pi + e$ or πe is irrational; both are expected to be transcendental. By contrast, e^π has been proved transcendental, and is therefore irrational.

5 Sets

Remark 5.1 (Membership and Classical Logic). [rem:membership-classical-logic] For every set A and every x in the chosen universe, exactly one of $x \in A$ and $x \notin A$ holds.

Proof. Let P be the proposition $x \in A$. By the law of excluded middle, $P \vee \neg P$, while the law of non-contradiction gives $\neg(P \wedge \neg P)$. Hence exactly one of $x \in A$ and $x \notin A$ holds. $\square$

Definition 5.1 (Intersection). [def:intersection] For sets A and B ,

$$x \in A \cap B \iff x \in A \text{ and } x \in B.$$

Definition 5.2 (Union). [def:union] For sets A and B ,

$$x \in A \cup B \iff x \in A \text{ or } x \in B.$$

Definition 5.3 (Complement). [def:complement] Given a universal set U and $A \subseteq U$, the complement of A is $A^c = U \setminus A$; equivalently,

$$x \in A^c \iff x \notin A.$$

Definition 5.4 (Set Difference). [def:set-difference] For sets A and B ,

$$x \in A \setminus B \iff x \in A \text{ and } x \notin B.$$

Definition 5.5 (Subset). [def:subset] For sets A and B ,

$$A \subseteq B \iff \text{for every } x, \quad x \in A \implies x \in B.$$

Definition 5.6 (Set Equality). [def:set-equality] For sets A and B ,

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Theorem 5.1 (De Morgan's Laws). [thm:de-morgan-laws] For sets A and B in a universal set U ,

$$(A \cap B)^c = A^c \cup B^c \quad \text{and} \quad (A \cup B)^c = A^c \cap B^c.$$

Also,

$$\left(\bigcap_{n=1}^{\infty} A_n \right)^c = \bigcup_{n=1}^{\infty} (A_n)^c \quad \text{and} \quad \left(\bigcup_{n=1}^{\infty} A_n \right)^c = \bigcap_{n=1}^{\infty} (A_n)^c$$

Example 5.1 (Nested Sequence). [eg:nested-sequence] Let $A_n = \{n, n+1, n+2, \dots\}, n \in \mathbb{N}$

ie, $A_1 = \{1, 2, 3, \dots\}, A_5 = \{5, 6, 7, \dots\}$

Trivially we can get $A_1 \supset A_2 \supset \dots$. This is called a nested sequence.

Where

$$\bigcup_{n=1}^{\infty} A_n = A_1 \quad \bigcap_{n=1}^{\infty} A_n = \emptyset$$

The intersection is $\emptyset$ due to if a number, let's say for any natural number p is in the set, then A_{p+1} does not include that number.

So in the universe $\mathbb{N}$,

$$\forall p \notin \bigcap_{n=1}^{\infty} A_n \implies \bigcap_{n=1}^{\infty} A_n = \emptyset$$

6 Mapping

Definition 6.1 (Mapping). [def:mapping] A mapping $f : A \rightarrow B$ is a set of rule st.:

$$\forall x \in A, \exists! y = f(x) \in B \leftrightarrow \forall x \in A, \exists! y \in B \text{ st. } (x, y) \in f$$

Definition 6.2 (Domain). [def:domain] The *domain* of a mapping $f : A \rightarrow B$ is the set A .

Definition 6.3 (Codomain). [def:codomain] The *codomain* of a mapping $f : A \rightarrow B$ is the set B .

Definition 6.4 (Range). [def:range] The *range* of a mapping $f : A \rightarrow B$ is a set C such that:

$$C = \{y : \exists x \in A, \text{ st. } y = f(x)\}$$

Remark 6.1 (Range and codomains are different). [rem:range-and-codomains-are-different] The range is a subset of codomain. Notice that some of the values in the codomain is not reachable.

For example,

$$f = x^2 : \mathbb{R} \rightarrow \mathbb{R}$$

The codomain is $\mathbb{R}$ but the range is $\mathbb{R}^+ \cup \{0\}$.

Definition 6.5 (Injective Mapping/One to One Mapping). [def:injective-mapping] A mapping $f : A \rightarrow B$ is called *injective* or *one to one* if

$$f(x_1) \neq f(x_2) \implies x_1 \neq x_2$$

equivalently,

$$x_1 = x_2 \implies f(x_1) = f(x_2)$$

Definition 6.6 (Surjective Mapping/onto Mapping). [def:surjective-mapping] A mapping $f : A \rightarrow B$ is called *surjective* or *onto* if

$$\forall y \in B, \exists x \in A, \text{ st. } f(x) = y$$

equivalently,

$$\text{Range}(f) = B$$

Definition 6.7 (Bijective Mapping). [def:bijective-mapping] A mapping $f : A \rightarrow B$ is called *bijective* if it is both injective and surjective.

7 Images

Definition 7.1 (Image). [def:image] For a mapping $f : A \rightarrow B$, we have set $C \subseteq A$. We define *image* as

$$f(C) := \{y : f(x), x \in C\} \quad \text{where} \quad f(C) \subseteq B$$

Definition 7.2 (Pre-Image). [def:pre-image] For a mapping $f : A \rightarrow B$, we have set $D \subseteq B$. We define *pre-image* as

$$f^{-1}(D) := \{x : f(x) \in D\} \quad \text{where} \quad f^{-1}(D) \subseteq A$$

Remark 7.1 (Image of Pre-image). [rem:image-of-pre-image] In general cases, for a mapping $f : A \rightarrow B$,

$$\begin{aligned} C \subseteq A, & \not\Rightarrow f^{-1}(f(C)) = C \\ D \subseteq B, & \not\Rightarrow f(f^{-1}(D)) = D \end{aligned}$$

Remark 7.2 (What to do when a result is not true?). [rem:how-to-treat-false-results] When a result such as the equalities in remark 7.1 is not true, we may:

- strengthen the assumptions, as in proposition 7.2;
- narrow down the conclusion, as in proposition 7.1.

Pf. The Subset Relation for Pre-image of Image. preimage-image-subset TODO (Counterexample) □

Proposition 7.1 (Subset Relation for Pre-image of Image). [prop:preimage-image-subset] In general cases, for a mapping $f : A \rightarrow B$,

$$\begin{aligned} C \subseteq A, & f^{-1}(f(C)) \supseteq C \\ D \subseteq B, & f(f^{-1}(D)) \subseteq D \end{aligned}$$

Proposition 7.2 (Pre-image of Image for Bijection Mapping). [prop:bijection-preimage-image] If $f : A \rightarrow B$ is a bijection mapping, then

$$\begin{aligned} C \subseteq A, & \implies f^{-1}(f(C)) = C \\ D \subseteq B, & \implies f(f^{-1}(D)) = D \end{aligned}$$

8 The Real Line

Definition 8.1 (Distance on the Real Line). [def:distance-on-the-real-line] For any two real numbers x and y , the *distance* between them is $|x - y|$.

Lemma 8.1 (ε -Criterion for Equality). [lem:epsilon-equality] For $a, b \in \mathbb{R}$,

$$a = b \iff \forall \varepsilon > 0, \quad |a - b| < \varepsilon.$$

Proof. In order to proof `lem:epsilon-equality`, we need to prove both directions.

(a): $a = b \implies \forall \varepsilon > 0, |a - b| < \varepsilon$

Suppose $a = b$. Then $|a - b| = 0$. For any $\varepsilon > 0$, we have $|a - b| = 0 < \varepsilon$.

(b): $\forall \varepsilon > 0, |a - b| < \varepsilon \implies a = b$

Assume $a \neq b$.

Then, this means that

$$\exists \varepsilon_0 > 0, |a - b| = \varepsilon_0$$

Taking $\varepsilon = \varepsilon_0$ gives $\varepsilon_0 = |a - b| < \varepsilon_0$, a contradiction.

So, we have $\forall \varepsilon > 0, |a - b| < \varepsilon \implies \neg a \neq b$

Since $\neg a \neq b \iff a = b$, So, $\forall \varepsilon > 0, |a - b| < \varepsilon \implies a = b$

Therefore, both implications hold, and hence proven. $\square$

9 Bounds

Definition 9.1 (Bounded Above). `[def:bounded-above]` A set A is *bounded above* in the universe U if

$$\exists b \in U, \forall a \in A, a \leq b$$

Such b is called an *upper bound* of A .

Definition 9.2 (Bounded Below). `[def:bounded-below]` A set A is *bounded below* in the universe U if

$$\exists b \in U, \forall a \in A, a \geq b$$

Such b is called a *lower bound* of A .

Definition 9.3 (Least Upper Bound/Supremum). `[def:supremum]` s is the *least upper bound* or *supremum* of a set A if

1. s is an upper bound of A
2. $\forall b$ is an upper bound of A , $s \leq b$

Proposition 9.1 (ε -Criterion for the Supremum). `[prop:epsilon-supremum]` Let $A \subseteq \mathbb{R}$ be nonempty and let $s \in \mathbb{R}$. Then $s = \sup A$ if and only if

1. s is an upper bound of A ; and
2. for every $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound of A .

Equivalently,

$$s = \sup A \iff \begin{cases} \forall x \in A, & x \leq s, \\ \forall \varepsilon > 0, & \exists x \in A \text{ s.t. } s - \varepsilon < x. \end{cases}$$

Proof. In order to proof ε -Characterization of the Supremum, we need to proof both directions are true.

(a):

$$s = \sup A \implies \begin{cases} \forall x \in A, & x \leq s, \\ \forall \varepsilon > 0, & \exists x \in A \text{ s.t. } s - \varepsilon < x. \end{cases}$$

According to the definition of supremum, we know that s is an upper bound of A , ie, $\forall x \in A, x \leq s$.

(b):

$$\begin{cases} \forall x \in A, & x \leq s, \\ \forall \varepsilon > 0, & \exists x \in A \text{ s.t. } s - \varepsilon < x. \end{cases} \implies s = \sup A$$

$\square$

Definition 9.4 (Greatest Lower Bound/Infimum). `[def:infimum]` i is the *greatest lower bound* or *infimum* of a set A if

1. i is a lower bound of A
2. $\forall b$ is a lower bound of A , $i \geq b$

Proposition 9.2 (ε -Criterion for the Infimum). [prop:epsilon-infimum] Let $A \subseteq \mathbb{R}$ be nonempty and let $i \in \mathbb{R}$. Then $i = \inf A$ if and only if

1. i is a lower bound of A ; and
2. for every $\varepsilon > 0$, $i + \varepsilon$ is not a lower bound of A .

Equivalently,

$$i = \inf A \iff \begin{cases} \forall x \in A, & i \leq x, \\ \forall \varepsilon > 0, & \exists x \in A \text{ s.t. } x < i + \varepsilon. \end{cases}$$

Convention 9.1. We use the notation $\sup A$ to denote the supremum of a set A and $\inf A$ to denote the infimum of a set A .

Remark 9.1 (Supremums and Infimums are unique). [rem:supremum-infimum-uniqueness] If a set A has a supremum, then it is unique. If a set A has an infimum, then it is unique.

Pf. Supremums are unique. Assume there are 2 supremums for set A , s_1, s_2 . Where $s_1 \neq s_2$.

Since $\forall s$ is an upper bound of A , $s_1 \leq s$, also $s_2 \leq s$.

Notice that s_1 is an upper bound of A , and s_2 is a least upper bound, so $s_2 \leq s_1$.

Also, s_2 is an upper bound of A , and s_1 is a least upper bound, so $s_1 \leq s_2$.

Trivially, $s_1 = s_2$. Contradicts with $s_1 \neq s_2$.

So there are no multiple supremum. Supremums are unique for each set. □

Definition 9.5 (Bounded Set). [def:bounded-set] A set A is *bounded* in the universe U if it is both bounded above and bounded below.

Definition 9.6 (Maximum). [def:maximum] A have a maximum $m = \max A$ if

$$\sup A \in A$$

Definition 9.7 (Minimum). [def:minimum] A have a minimum $m = \min A$ if

$$\inf A \in A$$

10 Axiom of Completeness

Definition 10.1 (Complete Ordered Field). [def:complete-ordered-field] A *complete ordered field* is a field F that is an ordered set and Every nonempty set $A \subseteq F$ that is bounded above has a least upper bound in F

Axiom 10.1 (Axiom of Completeness). [axm:aoc] Every nonempty set $A \subseteq \mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$

Lemma 10.1 (Supremum–Infimum Equivalence). [lem:supremum-infimum-equivalence] Let F be an ordered field. The following statements are equivalent:

1. Every nonempty set $A \subseteq F$ that is bounded above in F has a supremum in F .
2. Every nonempty set $A \subseteq F$ that is bounded below in F has an infimum in F .

Consequently, the definition of a complete ordered field may equivalently use the infimum property. In particular, the Axiom of Completeness is equivalent to the assertion that every nonempty subset of $\mathbb{R}$ bounded below has an infimum in $\mathbb{R}$.

Proof. First assume the supremum property, and let $A \subseteq F$ be nonempty and bounded below. Define

$$-A = \{-a : a \in A\}.$$

Then $-A$ is nonempty and bounded above, so let $s = \sup(-A)$. For every $a \in A$, we have $-a \leq s$, and hence $-s \leq a$. Thus $-s$ is a lower bound of A . If ℓ is any lower bound of A , then $-\ell$ is an upper bound of $-A$. Therefore $s \leq -\ell$, so $\ell \leq -s$. Hence $-s = \inf A$.

Conversely, assume the infimum property, and let $A \subseteq F$ be nonempty and bounded above. Then $-A$ is nonempty and bounded below, so let $i = \inf(-A)$. For every $a \in A$, we have $i \leq -a$, and hence $a \leq -i$. Thus $-i$ is an upper bound of A . If u is any upper bound of A , then $-u$ is a lower bound of $-A$. Therefore $-u \leq i$, so $-i \leq u$. Hence $-i = \sup A$. $\square$

Proposition 10.1 (The Rational Numbers Are Not Complete). [prop:rational-not-complete] *The ordered field $\mathbb{Q}$ is not complete. For example, the set*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}$$

is nonempty and bounded above in $\mathbb{Q}$, but it has no supremum in $\mathbb{Q}$.

Remark 10.1 (Integers and Natural Numbers). [rem:integers-naturals-not-fields] Neither $\mathbb{Z}$ nor $\mathbb{N}$ is a field, so neither can be called a complete ordered field. As ordered sets, however, their nonempty subsets that are bounded above have greatest elements.

Example 10.1. If $A = \{\frac{1}{n} : n \in \mathbb{N}\}$, then $\sup A = 1$ and $\inf A = 0$. Although it has a infimum, it does not have a minimum since $0 \notin A$.

Corollary 10.1 (Supremum and Infimum Existence in $\mathbb{R}$). [cor:supremum-infimum-existence] *Every nonempty set $A \subseteq \mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$ and every nonempty set $A \subseteq \mathbb{R}$ that is bounded below has a greatest lower bound in $\mathbb{R}$*

Proof. The supremum statement is the Axiom of Completeness, and the infimum statement follows from lemma 10.1. $\square$

Remark 10.2 (Intuition for Completeness). [rem:intuition-for-completeness] We can have a set, let's say,

$$S = \{r \in \mathbb{Q} : r^2 < 3\}, \text{ where } r \subset \mathbb{Q} \subset \mathbb{R}$$

Although the set S is bounded above in $\mathbb{Q}$, it does not have a supremum in $\mathbb{Q}$. According to the AoC, we know that $\exists s \in \mathbb{R}, s = \sup S$. This means we can find a number that can be described as the least upper bound a subset of rational numbers which is not in the rational numbers.